

Product Guide Specification

SECTION 09 97 00

SPECIAL COATINGS

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Special coating for concrete tanks and reservoirs used for the containment of potable water. Special coating rapidly cures to form a seamless, abrasion resistant, and potable water approved waterproof liner.

1.2 RELATED SECTIONS

- A. Section 03 30 00 –Cast-in-Place Concrete.
- B. Section 09 96 00 – High-Performance Coatings.
- C. Section 09 97 13 –Steel Coating.
- D. Section 09 97 23 –Concrete and Masonry Coating.
- E. Section 33 16 00 –Water Utility Storage Tank.
- F. Section 33 47 13.53 – Reservoir Liners.

1.3 REFERENCES

- A. ACI 201.1R – Guide for Making a Condition Survey of Concrete in Service.
- B. ANSI/NSF 61 – Drinking Water Systems Components – Health Effects.
- C. ASTM C836 – High-Solids Content, Cold Liquid-Applied Elastomeric Waterproofing Membrane for Use with Separate Wearing Course.
- D. ASTM C957 – High Solids Content, Cold-Liquid-Applied Elastomeric Waterproofing Membrane with Integral Wearing Surface.
- E. ASTM D36 – Softening Point of Bitumen (Ring-and-Ball Apparatus).

- F. ASTM D412 – Vulcanized Rubber and Thermoplastic Rubbers and Thermoplastic Elastomers – Tension.
- G. ASTM D624 – Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers.
- H. ASTM D648 – Deflection Temperature of Plastics under Flexural Load in the Edgewise Position.
- I. ASTM D751 – Coated Fabrics.
- J. ASTM D822 – Conducting Tests on Paint and Related Coatings and Materials using Filtered Open-Flame Carbon-Arc Light and Water Exposure Apparatus.
- K. ASTM D1117 – Evaluation Nonwoven Fabric.
- L. ASTM D1682 – Breaking Load and Elongation of Textile Fabric.
- M. ASTM D2240 – Rubber Property – Durometer Hardness.
- N. ASTM D3786 – Hydraulic Bursting Strength of Knitted Goods and Nonwoven Fabrics: Diaphragm Bursting Strength Tester Method.
- O. ASTM D4060 – Abrasion Resistance of Organic Coatings by the Taber Abraser.
- P. ASTM D4258 – Surface Cleaning Concrete for Coating.
- Q. ASTM D4259-88 – Abrading Concrete.
- R. ASTM D4263 – Indicating Moisture in Concrete by the Plastic Sheet Method.
- S. ASTM D4285 – Indicating Oil or Water in Compressed Air.
- T. ASTM E96 – Water Vapor Transmission of Materials.
- U. ICRI 03732 – Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, and Polymers Overlays.
- V. SSPC-SP10/NACE No. 2 – Near-White Blast Cleaning.
- W. SSPC-SP6/NACE No. 3 – Commercial Blast Cleaning.

1.4 SUBMITTALS

- A. Comply with requirements of Section 01 33 00 – Submittal Procedures.
- B. Product Data: Submit manufacturer’s product data, including surface preparation, application, and curing.

- C. Samples: Submit 3-inch by 1-inch samples for approval by the Engineer.
 - 1. Cured special coating, 25 mils thick.
 - 2. Reinforcing fabric and joint cover sheet.
- D. Potable Water Certification: Submit certification that coating has been classified by ANSI certified laboratory to ANSI/NSF 61 in tanks, reservoirs, pipes and joints.
- E. Applicator's Project References: Submit list of completed project references.
- F. Certification of Applicator's Supervisor: Submit for applicator's supervisor a certificate indicating completion of manufacturer's contractor training program.
- G. Warranty: Submit manufacturer's standard warranty.

1.5 QUALITY ASSURANCE

- A. Qualifications:
 - 1. Applicator: Use applicator experienced in the application of the specified special coating for a minimum of 2-years on projects of similar size and complexity. Provide a list of completed projects including project name and location, name of engineer, name of coating manufacturer, and approximate quantity of coating applied.
 - 2. Applicator's Supervisor: Employ a supervisor during all phases of the work that had successfully completed manufacturer's contractor training program.
 - 3. Applicator's Personnel: Employ persons trained for the application of special coating.
- B. Regulatory Requirements: Comply with environmental regulations.
- C. Pre-Application Meeting:
 - 1. Convene a pre-application meeting two weeks before the start of application of the special coating.
 - 2. Require attendance of parties directly affecting work of this section, including the contractor, sub-contractor, engineer, applicator, and manufacturer's representative.
 - 3. Review environmental requirements, materials, protection of adjacent work, surface preparation, application, curing, field quality control, cleaning, and coordination with other work.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Delivery:

1. Deliver materials to the site in manufacturer's original, unopened containers and packaging, with labels clearly indicating manufacturer and material.
2. Do not deliver material to site more than one month before use.

B. Storage:

1. Store the material in accordance with manufacturer's instructions.
2. Store materials indoor in an area well ventilated and protected from damage.
3. Do not store material near open flame, sparks, or hot surfaces.
4. Store materials on raised platforms and covered by waterproofing covers.
5. Keep material containers closed.

C. Handling: Protect materials during handling and application to prevent damage.

1.7 ENVIRONMENTAL REQUIREMENTS

- A. Do not apply in wet weather or when rain is imminent.
- B. Apply when the surface is a minimum 10 degrees C) and a minimum of 5 3 degrees C above dew point. Consult manufacturer for application instructions if the ambient or surface temperature is below 10 degrees F).
- C. Do not apply to porous substrates when substrate or ambient temperatures are rising.
- D. Do not apply to porous substrates when substrate is in direct sunlight.
- E. Do not apply over substrates that are frozen or contain frost.

1.8 WARRANTY

- A. Provide a 5-year material and 1-year labor warranty. Obtain material warranty from manufacturer.

PART 2 PRODUCTS

2.1 MANUFACTURER

- A. Saudi Vetosit Co. Ltd.-Saveto. P.O.Box 52235 - Riyadh 11563 Saudi Arabia. Tel: +966 11 265 3334 .Fax: +966 11 265 0936 ,www.saveto.com

2.2 SPECIAL COATINGS

- A. Special Coating: VETOPROOF EC722 is a two component Polysulphide modified epoxy protective coating system for concrete and steel surfaces consisting of a flexible primer EP722 and a flexible coating EC722.applied by brush, roller or airless spray application

1. Potable Water Service: Classified for potable water contact in tanks, reservoirs, pipes, and joints in accordance with ANSI/NSF 61
 2. Elongation at 7 days ASTM D 412, ASTM C836 and C957: %65
 3. Solids by volume: %100
 4. Hardness Shore D ASTM D2240: 40
 5. Tensile Strength ASTM C307: 5,690 psi
 6. Tensile Strength at 7 days ASTM D412: 13 N/mm²
- B. Primer: Vetoproof EP722: A two component solvent free flexible epoxy primer. Use as a primer coat on dry, porous substrates such as concrete.
1. Potable Water Service: Classified for potable water contact in tanks, reservoirs, pipes, and joints in accordance with ANSI/NSF 61.
 2. Solids by Volume: %100 mixed
 3. Elongation at 7 days ASTM D 412:%65
 4. Tensile Strength at 7 days ASTM D412: 14 N/mm²
- C. Patching Material: Vetorep ER350: A two component epoxy based thixotropic solvent free compound. Once components of the kit are mixed the products becomes a smooth paste that is used to repair surface imperfections and irregularities fill blow holes and pores while being completely impermeable and having excellent adhesive and bonding properties, the thixotropic nature of the products allows it to function in vertical and overhead situation without sagging while maintaining excellent bond.
1. Potable Water Service: Classified for potable water contact in tanks, reservoirs, pipes, and joints in accordance with ANSI/NSF 61
 2. Solids by volume: %100
 3. Compressive Strength ASTM C579, 7 days): 55N/mm².
 4. Tensile Strength ASTM C579, 7 days: 12 N/mm²
 5. Flexural Strength BS 6319, Pt3: 23 MPa @ 28 Days
 6. VOC Content ASTM D2369: <10 gm < Liter.

PART 3 EXECUTION

3.1 INSPECT

- A. Inspect substrate and adjacent areas where special coating will be applied. Notify the Engineer of conditions that would adversely affect the application or subsequent utilization of the special coating. Do not proceed with application until unsatisfactory conditions are corrected.

3.2 PROTECTION

- A. Protect adjacent work and surrounding areas from contact with special coating.

3.3 SURFACE PREPARATION FOR CONCRETE TANKS AND RESERVOIRS

- A. Prepare surface in accordance with manufacturer's instructions.
- B. Provide clean, dry, and structurally sound concrete surface.
- C. New Concrete:
 - 1. Ensure concrete has a minimum compressive strength of 3,000 psi, is dry, and is free of release agents and curing compounds before application of special coating.
 - 2. Remove surface laitance and expose the underlying aggregate consistent with ICRI CSP 4 to 6 in accordance with ICRI 03732.
- D. Existing Concrete: Use existing coating as support for the special coating. Patch existing coating as approved by the Engineer. Abrade the existing coating and apply sample patch to test for suitability and adhesion.]
- E. Condition Survey: Perform a condition survey of existing concrete in accordance with ACI 201.1R.
- F. Abrasive Blasting:
 - 1. Prepare concrete surface to receive special coating by abrasive blasting.
 - 2. Remove dirt, soil, grease, oil, paint, coatings, form release agents, curing compounds, laitance, loose material, unsound concrete, and other foreign materials that would inhibit performance of special coating in accordance with ASTM D4258 and by abrasive blasting.
 - 3. Obtain a firm, sound concrete surface in which bug holes are fully opened or repaired.
 - 4. Remove sharp concrete edges and projections.
 - 5. Perform abrasive blasting in accordance with ASTM D4259-88.
 - 6. Receive approval by Engineer of blasting media.
 - 7. Maintain air supply for abrasive blasting free of oil and water in accordance with ASTM D4285.
 - 8. Expose aggregate to obtain a profile of ICRI CSP 4 to 6 in accordance with ICRI 03732.
- G. Repair concrete surface to be free of holes. Fully open bug holes before repair. Repair defects in the concrete surface, such as bug holes, air pockets, and honeycomb by filling and smoothing off with patching material, epoxy patching compound, or grout. Abrasive blast repaired surfaces.
- H. Ensure substrate is clean and dry in accordance with manufacturer's instructions. Remove surface laitance from concrete surface to expose aggregate to obtain a profile of ICRI CSP 4 to 6 in accordance with ICRI 03732.
- I. Repair cracks in concrete surface with material suitable for type and width of crack, compatible with substrate and special coating, and approved by the Engineer.

- J. Moisture Tests: Do not apply primer or special coating to concrete surface unless two or more of the following moisture tests confirm appropriate moisture levels for properly prepared substrates:
1. Plastic Sheet Method (ASTM D4263): Pass/Fail.
 2. Relative Humidity Test: Less than 75 percent relative humidity at 70 degrees F.
 3. Calcium Chloride Test: Less than 5 pounds per 1,000 square feet per 24 hours.
 4. Radio Frequency Test: Less than 5 percent moisture.

3.4 APPLICATION

- A. Apply primer to concrete surface a minimum of 4-mils wet thickness. A Uniform coating free of holidays or pinholes is necessary to minimize outgassing effects during the application of the special coating to porous surfaces such as concrete. Surfaces may require additional coats to obtain a pinhole free finish.
- B. Allow primer to cure in accordance with manufacturer's instructions before top coating with the special coating.
- C. Apply Vetoproof EC722 in accordance with manufacturer's instructions.
- D. Keep material containers tightly closed until ready for use.
- E. Keep equipment, air supplies, and application surfaces dry.
- F. Mix and apply when special coating is above 15 degrees C.
- G. Do not use adulterants, thinners, or cutback solutions.
- H. Blend and mix 2-component materials in accordance with manufacturer's instructions. Do not hand mix components.
- I. Maintain air supply for material spray application free of oil and water in accordance with ASTM D4285.
- J. Apply special coating directly to a clean and dry surface or to reinforcing fabric.
- K. Apply a 6 to 12-inch wide strip of joint cover sheet over cracks over 1/8-inch wide, non-working joints, and edges. Adhere center joint cover sheet over all joints by applying a tack coat of the special coating.
- L. Apply sufficient special coating to achieve 60-mils wet film thickness for containment of potable water.
- M. Joint Lines:
1. Prepare for joint lines should rain or other conditions require work stoppage or extended delay.

2. Install joint lines clean and straight. Install overlap 6-inches minimum to ensure an impervious joint.
3. Severely abrade with wire brush or sandpaper and apply bonding agent to all areas where the special coating has cured beyond its recoat window.

N. Recoating:

1. Recoat the special coating system within the recoat window to obtain maximum interlayer adhesion to build specific thickness.
2. Immersion Service: Minimize areas to be recoated outside the recoat window, except at joint lines.
3. Non-Immersion Service: Severely abrade with wire brush or surface grinder, apply bonding agent, and recoat, if special coating has cured more than the recoat window. Acceptable adhesion can only be achieved through aggressive abrading.

3.5 CURING

- A. Cure special coating in accordance with manufacturer's instructions.
- B. Curing Time: Allow sufficient time for solvents to evaporate from the cured special coating before placing into service.
- C. Receive approval of cured coating by Engineer.

3.6 FIELD QUALITY CONTROL

- A. Provide inspection services by an independent inspection firm throughout all phases of surface preparation, application, and curing of the special coating.

3.7 CLEANING

- A. Remove and dispose of all temporary materials used to protect adjacent work and surrounding areas.
- B. Immediately remove and clean special coating materials from surfaces not intended to receive the materials.

END OF SECTION